

KEMI TORNIO, Finland

WMO: 028640

Lat: 65.784N

Lon: 24.579E

Elev: 19

StdP: 101.10

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-24.9	-30.9	0.2	-27.7	-27.6	0.3	-24.8	11.3	-2.2	10.2	-2.1	1.6	10	0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	24.8	17.6	22.9	16.9	21.1	16.0	19.2	23.0	17.9	21.4	16.8	19.9	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	21.1	16.4	11.7	19.4	15.3	10.9	18.2	54.5	22.7	50.5	21.6	47.2	20.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.4	DB	-30.5	27.0	3.3	2.5	-32.9	28.8	-34.8	30.2	-36.7	31.6	-39.0	33.4	(4)
(5)			WB	-30.7	20.2	3.2	1.9	-33.0	21.6	-34.9	22.7	-36.7	23.7	-39.0	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-10.2	-9.9	-5.7	0.6	6.7	12.0	15.8	13.6	9.0	2.5	-2.8	-6.1	(6)	
	DBStd	10.15	7.48	7.48	5.40	3.32	4.15	3.25	3.05	3.26	3.46	4.21	5.52	6.63	(7)	
	HDD10.0	3271	626	557	488	283	119	16	1	8	58	234	383	499	(8)	
	HDD18.3	5906	884	790	746	532	361	193	89	149	279	492	633	757	(9)	
	CDD10.0	423	0	0	0	0	18	74	181	120	29	1	0	0	(10)	
	CDD18.3	17	0	0	0	0	1	2	11	3	0	0	0	0	(11)	
	CDH23.3	134	0	0	0	0	11	19	90	14	0	0	0	0	(12)	
	CDH26.7	19	0	0	0	0	1	4	14	1	0	0	0	0	(13)	
(14) Wind	WSAvg	3.3	3.2	3.3	3.4	3.3	3.5	3.5	3.1	3.2	3.4	3.4	3.4	3.4	(14)	
(15) Precipitation	PrecAvg	602	51	41	35	30	40	46	70	61	54	57	62	51	(15)	
	PrecMax	838	92	92	63	81	69	80	133	151	127	108	111	120	(16)	
	PrecMin	457	12	4	10	7	11	9	10	13	15	20	18	15	(17)	
	PrecStd	117	23	23	15	20	16	22	32	36	30	24	26	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.4	4.7	6.0	12.9	24.9	26.2	28.8	25.2	18.5	11.8	7.6	4.3	(19)	
		MCWB	1.6	2.2	2.2	6.7	14.5	17.9	19.4	18.2	14.2	9.9	6.7	3.5	(20)	
	2%	DB	1.2	2.2	4.0	9.8	20.1	22.2	25.7	22.8	16.6	10.5	6.6	3.2	(21)	
		MCWB	0.8	1.2	1.6	5.5	12.4	15.3	18.1	17.2	13.3	9.3	6.1	2.6	(22)	
	5%	DB	0.5	0.9	2.2	7.3	16.5	20.1	23.7	21.0	15.3	9.5	5.1	2.2	(23)	
		MCWB	0.3	0.5	0.7	3.9	10.6	13.8	17.6	16.5	13.0	8.4	4.6	1.7	(24)	
	10%	DB	-0.2	0.0	1.1	5.5	13.9	18.1	22.0	19.3	14.3	8.2	4.0	1.5	(25)	
		MCWB	-0.4	-0.3	0.1	2.7	9.3	13.0	16.9	15.9	12.5	7.3	3.5	1.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	2.7	3.0	8.1	15.4	18.9	21.1	20.4	15.4	10.7	7.1	3.8	(27)	
		MCDB	2.0	4.1	5.0	10.6	22.4	24.0	25.0	23.3	17.2	11.3	7.3	4.1	(28)	
	2%	WB	0.9	1.3	1.8	5.9	13.3	16.8	19.7	18.5	14.3	9.6	5.9	2.7	(29)	
		MCDB	1.2	2.0	3.1	9.1	19.1	21.1	23.5	21.2	15.7	10.3	6.3	3.0	(30)	
	5%	WB	0.2	0.5	1.0	4.2	11.3	15.2	18.6	17.3	13.6	8.5	4.6	1.8	(31)	
		MCDB	0.4	0.9	1.9	6.9	15.5	18.3	22.7	19.6	15.0	9.2	4.9	2.2	(32)	
	10%	WB	-0.5	-0.3	0.3	3.1	9.7	13.8	17.5	16.3	12.8	7.4	3.5	1.0	(33)	
		MCDB	-0.3	0.0	1.0	5.0	13.1	16.8	21.1	18.7	14.2	8.1	3.9	1.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.8	8.8	7.6	8.8	8.4	9.0	8.3	7.3	5.8	5.4	6.3	(35)	
		MCDBR	5.7	5.7	7.5	10.2	13.4	12.1	12.2	10.4	8.3	6.1	4.7	4.8	(36)	
	5% WB	MCWBR	5.6	5.2	5.8	6.8	7.3	6.1	6.4	6.0	5.8	5.2	4.7	4.6	(37)	
		MCWBR	5.8	5.5	6.3	9.3	12.0	8.7	10.1	8.2	6.6	5.5	4.5	4.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.206	0.273	0.302	0.339	0.345	0.349	0.366	0.350	0.327	0.302	0.344	0.356	(40)	
		taud	1.944	2.095	2.151	2.214	2.422	2.478	2.451	2.483	2.528	2.347	2.573	2.612	(41)	
	Edn at Noon	Edn at Noon	245	580	753	819	859	864	834	812	752	564	237	76	(42)	
		Edh at Noon	19	57	88	107	97	94	94	83	63	46	18	9	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.08	0.56	2.03	3.76	4.91	5.50	5.12	3.73	1.99	0.71	0.14	0.02	(44)	
	RadStd	RadStd	0.01	0.11	0.21	0.30	0.45	0.42	0.45	0.34	0.21	0.12	0.02	0.00	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page